Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 1 / 方法 1 [LO03-M01]

A logical case split must preserve every condition and exhaust every possible model. Separate ordinary feasibility, ambiguity after partial information, and public knowledge: they remove possibilities in different ways.

逻辑分类必须保留所有条件，并覆盖全部可能模型。普通可行性、部分信息造成的歧义及公开知识更新，会以不同方式排除可能性，应分别处理。

Propagate the tightest local constraint before branching

先传播最紧的局部限制再分类

Procedure

1. Choose the smallest uncovered position, a forced repeated value, or a variable involved in many restrictions.

2. For each possible branch, immediately propagate all forced inclusions and exclusions.

3. Count only after proving that every remaining constraint has been satisfied or explicitly retained.

4. Use size bounds to close the search: k distinct positive integers sum to at least $k(k+1)/2$.

通用步骤

1. 选择最小未覆盖位置、被迫重复的值，或参与限制最多的变量。

2. 每个分支一旦确定，立刻传播所有必选和禁选条件。

3. 只有证明其余限制均已满足或仍被明确保留后，才能计数。

4. 用数量界封闭搜索：k 个不同正整数之和至少为 $k(k+1)/2$。

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Method 1 / 方法 1 [LO03-M01] (continued / 续)

> Example. A set of chosen tasks is selected from A, B, C, D, E. Choosing A forbids B, C, and choosing D requires C. How many task sets are possible?

> 例题：从任务 A, B, C, D, E 中选择子集。选择 A 就不能选 B, C，选择 D 则必须选 C。共有多少种可行任务集?

Worked solution

1. If A is chosen, B, C are excluded, hence D is excluded too. Only E is free, giving two sets.

2. If A is absent, B, E are free. The allowed (C, D) states are $(0,0), (1,0), (1,1)$, giving $2 \cdot 2 \cdot 3 = 12$ sets.

3. The two branches are disjoint and exhaustive, so the total is 14. The implication was enforced before the independent choices were multiplied.

例题解析

1. 若选 A，则 B, C 不选，由此 D 也不能选，只剩 E 自由，共两种。

2. 若不选 A，则 B, E 自由，(C, D) 只能取 $(0,0), (1,0), (1,1)$，共 $2 \cdot 2 \cdot 3 = 12$ 种。

3. 两类不相交且覆盖全部可能，共 14 种。先执行蕴含条件，才可把独立选择相乘。

A short list is exhaustive only after an explicit bound or forced first case has been proved.

只有先证明界或被迫的首个分类，短候选表才称得上穷尽。

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Method 1 / 方法 1 [LO03-M01] (continued / 续)

For a progression $r, r+d, r+2d, \ldots$ to cover an uncovered index q, its step must divide $q-r$ and $q \geq r$. To forbid overlap of two progressions, use their congruence compatibility modulo the gcd of their steps.

For an absent integer median, the list is even and its two middle entries are distinct with the same parity. A unique mode has strictly greater multiplicity than every other value; these constraints combine with the distinct-positive sum bound.

若数列 $r, r+d, r+2d, \ldots$ 要覆盖未覆盖位置 q，必须有 $d \mid (q-r)$ 且 $q \geq r$。判断两等差数列是否重叠，可检查它们的同余条件在公差最大公因数下是否相容。

若中位数为未出现的整数，则列表长为偶数，中间两项不同且同奇偶；唯一众数的频数严格超过每个其他值。这些限制应与不同正数的最小和界合用。

Practice / 配套练习: AIME-2002-I-Q09 (WB p. 496), AIME-2017-I-Q12 (WB p. 644), AIME-2024-II-Q02 (WB p. 795)

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Method 2 / 方法 2 [LO03-M02]

Group complete models by the information that is known

按已知信息把完整模型分组

Procedure

1. Enumerate the complete feasible models under a proved finite bound.

2. Compute the reported summary for each model, and group models having the same summary.

3. Translate “still two possibilities” into a group-size condition, then compute the requested quantity from the surviving group.

4. Distinguish distinct models from sign, order, or labeling copies that describe the same model.

通用步骤

1. 在已证明的有限范围内，列出完整可行模型。

2. 计算每个模型的已知摘要，按摘要相同分组。

3. 将“仍有两种可能”转为组大小条件，再由保留组求目标量。

4. 区分真正不同的模型与符号、顺序或标号变化造成的重复。

> Example. Three distinct members of $\{1, 2, 3, 4, 5, 6\}$ are chosen. Only their product is announced. Which products leave exactly two possible chosen triples?

> 例题：从 $\{1, 2, 3, 4, 5, 6\}$ 中选三个不同数，只公布乘积。哪些乘积仍对应恰好两种所选三元集？

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Method 2 / 方法 2 [LO03-M02] (continued / 续)

Worked solution

1. There are $\binom{6}{3} = 20$ complete models. Compare products for these unordered triples rather than their six orderings.

2. The repeated-product groups are $12 : \{1, 2, 6\}, \{1, 3, 4\}$; $24 : \{1, 4, 6\}, \{2, 3, 4\}$; $30 : \{1, 5, 6\}, \{2, 3, 5\}$; and $60 : \{2, 5, 6\}, \{3, 4, 5\}$.

3. The other twelve products are $6, 8, 10, 15, 18, 20, 36, 40, 48, 72, 90, 120$, each occurring once. Thus exactly $12, 24, 30, 60$ have the required ambiguity.

例题解析

1. 共有 $\binom{6}{3} = 20$ 个完整模型，应对无序三元集的乘积分组，而非重复计其六种顺序。

2. 重复组为 $12 : \{1, 2, 6\}, \{1, 3, 4\}$；$24 : \{1, 4, 6\}, \{2, 3, 4\}$；$30 : \{1, 5, 6\}, \{2, 3, 5\}$；$60 : \{2, 5, 6\}, \{3, 4, 5\}$。

3. 其余十二个乘积为 $6, 8, 10, 15, 18, 20, 36, 40, 48, 72, 90, 120$，各出现一次。因此所求恰为 $12, 24, 30, 60$。

An ambiguity condition refers to the entire feasible class, not merely two examples you happened to find.

歧义条件针对完整可行类，不能只找到两个例子就断言恰有两个。

Practice / 配套练习: AIME-2015-II-Q06 (WB p. 48)

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Method 3 / 方法 3 [LO03-M03]

Pair complementary sums of four unknown numbers

将四个未知数的两两和配成互补对

Procedure

1. For four numbers with total T, partition the six pair sums into three complementary pairs each totaling T.

2. Use the sum of all six entries, $3T$, and any known largest or smallest sums to restrict the pairing.

3. For each surviving pairing, solve three linear pair equations to recover the four numbers.

4. Check distinctness and reproduce all six pair sums to prove attainment.

通用步骤

1. 四个数总和为 T 时，六个两两和可配成三对，每对之和均为 T。

2. 利用六和总和 $3T$，以及已知最大、最小和来限制配对。

3. 对每种保留配对，解三个线性两两和方程，恢复四个数。

4. 检查数互异并重算全部六和，证明该配对可实现。

> Example. Four distinct real numbers have pair sums $4, 7, 8, 9, 10, 13$. Find all possible sorted quadruples.

> 例题：四个不同实数的六个两两和为 $4, 7, 8, 9, 10, 13$。求所有从小到大的四元组。

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Method 3 / 方法 3 [LO03-M03] (continued / 续)

Worked solution

1. The total of the four numbers is $T = (4+7+8+9+10+13)/3 = 17$. Complementary pairs are $(4,13),(7,10),(8,9)$.

2. For $a<b<c<d$, the smallest sums are $a+b=4, a+c=7$, and $c+d=13$. The remaining ambiguity is $a+d=8, b+c=9$ or the reverse.

3. The first gives $(a,b,c,d)=(1,3,6,7)$; the second gives $(3/2,5/2,11/2,15/2)$. Both are distinct sorted quadruples and reproduce exactly the six given sums, so the list is complete.

例题解析

1. 四数和为 $T = (4+7+8+9+10+13)/3 = 17$，互补对为 $(4,13),(7,10),(8,9)$。

2. 若 $a<b<c<d$，则最小两和为 $a+b=4, a+c=7$，最大和为 $c+d=13$。余下仅两种：$a+d=8, b+c=9$，或交换八与九。

3. 第一种给出 $(1,3,6,7)$，第二种给出 $(3/2,5/2,11/2,15/2)$。两组均严格递增，且重算得到全部六和，因此已穷尽。

Equal complementary totals are necessary; explicit reconstruction proves sufficiency.

互补对总和相等只是必要条件，恢复具体数值才能证明充分。

Practice / 配套练习: AIME-2017-II-Q05 (WB p. 18)

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Method 4 / 方法 4 [LO03-M04]

Update possible worlds after a public statement of ignorance

公开“不知道”后更新可能世界

Procedure

1. List all worlds allowed by the initial public information.

2. For each possible private observation, count how many initial worlds contain it; a unique world would make its observer know.

3. A simultaneous public statement that everyone is ignorant removes each world containing any initially unique observation.

4. Recompute observation classes among the surviving worlds and check the later statement, using the correct simultaneous round.

通用步骤

1. 先列出初始公开信息允许的全部可能世界。

2. 对每种私人观察，数它属于多少个初始世界；若只属于一个，观察者就能确定。

3. 若所有人同时公开表示不知道，就排除含有任何初始唯一观察的世界。

4. 在保留世界中重新分组私人观察，检查下一轮陈述，且必须遵守同时发言的轮次。

> Example. A teacher chooses one pair from $\{a,b\},\{b,c\},\{c,d\}$ and gives its two distinct symbols privately to two perfectly logical pupils. Both simultaneously say they cannot identify the pair. Which pair was chosen, and can both identify it afterward?

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Method 4 / 方法 4 [LO03-M04] (continued / 续)

> 例题：老师从 $\{a,b\},\{b,c\},\{c,d\}$ 中选择一对，把两个不同符号分别私下给两位完全理性的学生。两人同时表示不能确定所选对。选的是哪一对？听完之后两人能否确定？

Worked solution

1. Initially a occurs only in $\{a,b\}$ and d only in $\{c,d\}$, so either symbol would allow its pupil to know immediately.

2. Unanimous initial ignorance therefore excludes both pairs containing a or d, leaving only $\{b,c\}$.

3. After the public update, either private symbol b or c belongs to that sole surviving world. Both pupils can now identify the pair.

例题解析

1. 初始时，a 仅属于 $\{a,b\}$，d 仅属于 $\{c,d\}$，拿到它们的人会立即知道。

2. 两人都表示初始不知道，就排除了含 a 或 d 的两对，只剩 $\{b,c\}$。

3. 公开更新后，无论私人符号为 b 还是 c，都只对应这个保留世界，所以两人此时都能确定。

Analyze all initial replies against the same initial world set; processing simultaneous replies sequentially changes the puzzle.

同一轮初始回答必须都针对同一个初始世界集；把同时回答逐个更新会改变题意。

Practice / 配套练习: AIME-2021-II-Q11 (WB p. 810)